

7.3 Operational Properties I

In this section and the next one, we will introduce some very useful properties of the Laplace Transform.

What is $\mathcal{L}\{e^{3t} \cos(4t)\}$?

This is a special case of:

$$\mathcal{L}\{e^{at} f(t)\}.$$

observe that:

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt = F(s)$$

$$\begin{aligned}\mathcal{L}\{e^{at} f(t)\} &= \int_0^{\infty} e^{-st} e^{at} f(t) dt \\ &= \int_0^{\infty} e^{-(s-a)t} f(t) dt = F(s-a)\end{aligned}$$

1st shifting Theorem

If $\mathcal{L}\{f(t)\} = F(s)$, then:

$$\mathcal{L}\{e^{at} f(t)\} = F(s-a).$$

For example, we know that:

$$\begin{cases} \mathcal{L}\{\cos(\omega t)\} = \frac{s}{s^2 + \omega^2} \\ \mathcal{L}\{\sin(\omega t)\} = \frac{\omega}{s^2 + \omega^2} \\ \mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}} \end{cases}$$

$$\Rightarrow \begin{cases} \mathcal{L}\{e^{at} \cos(\omega t)\} = \frac{s-a}{(s-a)^2 + \omega^2} \\ \mathcal{L}\{e^{at} \sin(\omega t)\} = \frac{\omega}{(s-a)^2 + \omega^2} \\ \mathcal{L}\{e^{at} t^n\} = \frac{n!}{(s-a)^{n+1}} \end{cases}$$

Notation

$$\mathcal{L}\{e^{at} f(t)\} = \mathcal{L}\{f(t)\} \Big|_{s \rightarrow s-a}$$

$s \rightarrow s-a$ means that we replace s by $s-a$.

$$\begin{aligned} \mathcal{L}^{-1}\{F(s-a)\} &= e^{at} \mathcal{L}^{-1}\{F(s)\} \\ &= e^{at} f(t) \end{aligned}$$

Note that the denominators of $\mathcal{L}\{e^{at} \cos(\omega t)\}$ and $\mathcal{L}\{e^{at} \sin(\omega t)\}$ are $(s-a)^2 + \omega^2$.

This form can be obtained by completing the square of an irreducible quadratic.

How to deal with quadratics.

ex: ① Evaluate $\mathcal{L}^{-1}\{F(s)\}$

For : (a) $F(s) = \frac{3s+7}{s^2+4s+3}$

(b) $F(s) = \frac{3s+7}{s^2+4s+4}$

(c) $F(s) = \frac{3s+7}{s^2+4s+13}$

(a) s^2+4s+3 can be factored so we use Partial Fractions.

$$F(s) = \frac{3s+7}{(s+3)(s+1)} = \frac{A}{s+3} + \frac{B}{s+1}$$

Solve for A, B to get: $A=1$ and $B=2$.

$$\Rightarrow \mathcal{L}^{-1}\{F(s)\} = e^{-3t} + 2e^{-t}.$$

$$(b) \quad s^2 + 4s + 4 = (s+2)^2$$

$$F(s) = \frac{3s+7}{(s+2)^2} = \frac{3(s+2)+1}{(s+2)^2} = \frac{3}{s+2} + \frac{1}{(s+2)^2}$$

$$\mathcal{L}^{-1}\{F(s)\} = 3e^{-2t} + te^{-2t}$$

(c) $s^2 + 4s + 13$ is irreducible so we have to complete the square.

$$s^2 + 4s + 13 = (s^2 + 4s + 4) + 9 = (s+2)^2 + 9$$

$$\Rightarrow F(s) = \frac{3s+7}{(s+2)^2 + 9}$$

$$= \frac{3(s+2)+1}{(s+2)^2 + 9}$$

$$= 3 \left(\frac{s+2}{(s+2)^2 + 9} \right) + \frac{1}{(s+2)^2 + 9}$$

$$F(s) = 3 \left(\frac{s+2}{(s+2)^2 + 9} \right) + \underbrace{\frac{1}{3} \left(\frac{\textcircled{3}}{(s+2)^2 + 9} \right)}$$

multiply and divide by $\textcircled{3}$
to match the pattern

$$\frac{\omega}{(s-a)^2 + \omega^2}$$

$$\mathcal{L}^{-1}\{F(s)\} = 3e^{-2t} \cos(3t) + \frac{1}{3}e^{-2t} \sin(3t)$$

To summarize the technique for part (c), we complete the square and "massage" the numerator to match the patterns of:

$$\mathcal{L}\{e^{at} \cos(\omega t)\} = \frac{s-a}{(s-a)^2 + \omega^2}$$

and

$$\mathcal{L}\{e^{at} \sin(\omega t)\} = \frac{\omega}{(s-a)^2 + \omega^2}.$$

ex: ② Evaluate $\mathcal{L}^{-1} \left\{ \frac{4s+5}{(s-3)^4} \right\}$.

we could use P.F.

$$\frac{4s+5}{(s-3)^4} = \frac{A}{s-3} + \frac{B}{(s-3)^2} + \frac{C}{(s-3)^3} + \frac{D}{(s-3)^4}$$

but there's a much faster way.

$$\frac{4s+5}{(s-3)^4} = \frac{4(s-3)+5+12}{(s-3)^4}$$

$$= \frac{4(s-3)+17}{(s-3)^4}$$

$$= \frac{4}{(s-3)^3} + \frac{17}{(s-3)^4}$$

Note: the P.F. above would have

$$A=0, B=0, C=4, D=17.$$

$$\text{Since: } \mathcal{L}\{e^{at} t^n\} = \frac{n!}{(s-a)^{n+1}}$$

$$\Rightarrow \mathcal{L}^{-1}\left\{\frac{1}{(s-a)^{n+1}}\right\} = \frac{e^{at} t^n}{n!}$$

$$\mathcal{L}^{-1}\left\{\frac{4s+5}{(s-3)^4}\right\} = 4 \mathcal{L}^{-1}\left\{\frac{1}{(s-3)^3}\right\} + 17 \mathcal{L}^{-1}\left\{\frac{1}{(s-3)^4}\right\}$$

$$= 4 \left(\frac{t^2 e^{3t}}{2!} \right) + 17 \left(\frac{t^3 e^{3t}}{3!} \right)$$

$$= \boxed{2t^2 e^{3t} + \frac{17}{6} t^3 e^{3t}}$$

ex: ③ Solve the IVP

$$\begin{cases} y''(t) + 2y'(t) + 5y(t) = e^{-at} \\ y(0) = 1 \\ y'(0) = 2. \end{cases}$$

Take Laplace Transform on both sides of the DE.

$$\mathcal{L}\{y''(t)\} + 2\mathcal{L}\{y'(t)\} + 5\mathcal{L}\{y(t)\} = \mathcal{L}\{e^{-at}\}$$

$$\left[s^2 Y(s) - sy(0) - y'(0)\right] + 2\left[sY(s) - y(0)\right] + 5Y(s) = \frac{1}{s+a}$$

$$(s^2 + 2s + 5)Y(s) - s - 2 - 2 = \frac{1}{s+a}$$

$$\Rightarrow Y(s) = \frac{s+4}{s^2+2s+5} + \frac{1}{(s+a)(s^2+2s+5)}$$

$s^2 + 2s + 5$ is an irreducible quadratic.

Let's use P.F. for $\frac{1}{(s+2)(s^2+2s+5)}$

$$\frac{1}{(s+2)(s^2+2s+5)} = \frac{A}{s+2} + \frac{Bs+C}{s^2+2s+5}$$

$$1 = A(s^2+2s+5) + (Bs+C)(s+2)$$

$$s = -2 \rightarrow 1 = 5A + 0 \rightarrow A = 1/5$$

$$s = 0 \rightarrow 1 = 5A + 2C \rightarrow C = 0$$

$$s = 1 \rightarrow 1 = 8A + 3B + 3C \rightarrow B = \frac{1-8A}{3} = -1/5$$

$$Y(s) = \frac{s+4}{s^2+2s+5} + \frac{1/5}{s+2} + \frac{(-1/5)s+0}{s^2+2s+5}$$

$$Y(s) = \frac{1/5}{s+2} + \frac{\frac{4}{5}s+4}{s^2+2s+5}$$

Complete the square

$$\begin{aligned} s^2 + 2s + 5 &= (s^2 + 2s + 1) + 4 \\ &= (s+1)^2 + 4 \end{aligned}$$

$$Y(s) = \frac{1/5}{s+2} + \frac{\frac{4}{5}s + 4}{(s+1)^2 + 4}$$

$$Y(s) = \frac{1/5}{s+2} + \frac{\frac{4}{5}(s+1) + 4 - 4/5}{(s+1)^2 + 4}$$

$$Y(s) = \frac{1/5}{s+2} + \frac{\frac{4}{5}(s+1) + \frac{16}{5}}{(s+1)^2 + 4}$$

$$Y(s) = \frac{1}{5} \left(\frac{1}{s+2} \right) + \frac{4}{5} \left(\frac{s+1}{(s+1)^2 + 4} \right) + \frac{8}{5} \left(\frac{2}{(s+1)^2 + 4} \right)$$

$$y(t) = \mathcal{L}^{-1}\{Y(s)\}$$

$$\Rightarrow y(t) = \frac{1}{5} e^{-2t} + \frac{4}{5} e^{-t} \cos(2t) + \frac{8}{5} e^{-t} \sin(2t)$$

Discontinuities and Step Functions

In Science and Engineering, there are frequently functions that are either "on" or "off".

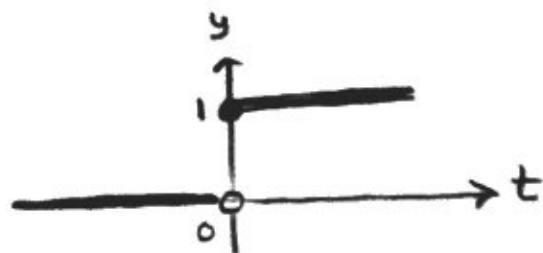
For example a voltage can be turned off after a period of time.

We will now define a special function that is 0 (off) if $t < 0$ and 1 (on) if $t \geq 0$.

$$u(t) = \begin{cases} 1 & \text{if } t \geq 0 \\ 0 & \text{if } t < 0 \end{cases}$$

$u(t)$ is called a unit step function

or Heaviside function.

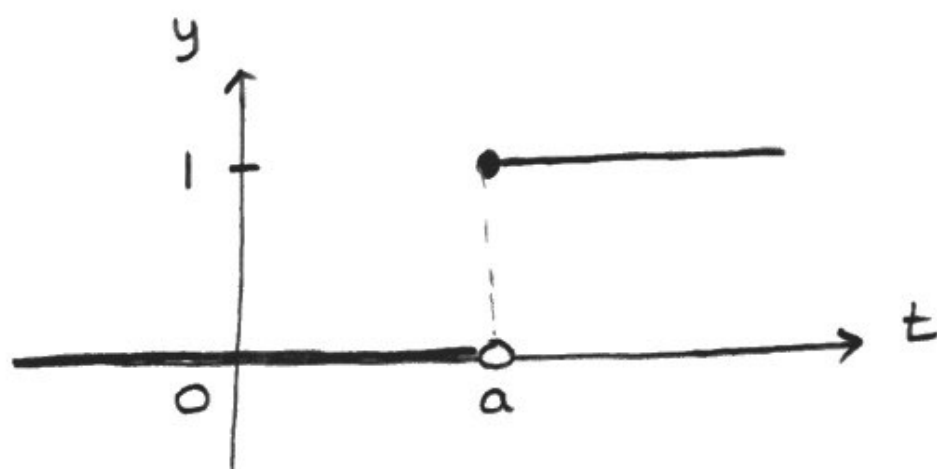


$$y = u(t)$$

The function $u(t-a)$ is

"off" for $t < a$ and "on" for $t \geq a$.

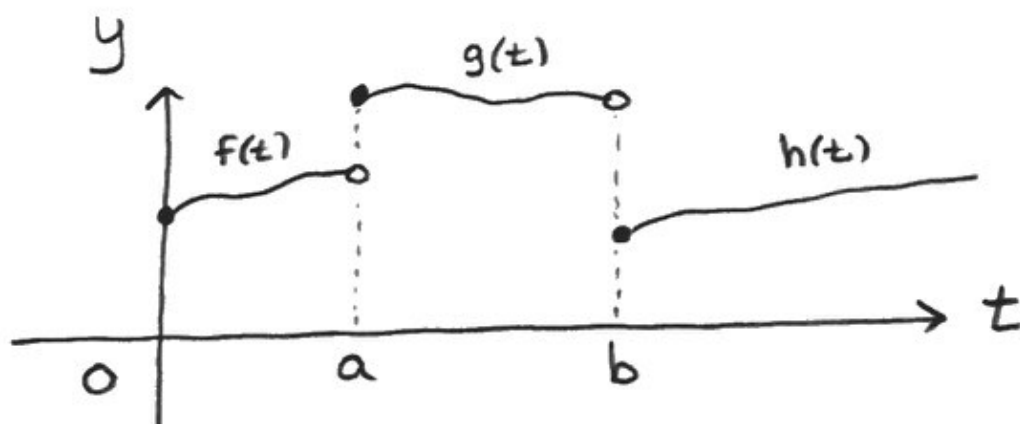
$$u(t-a) = \begin{cases} 1 & \text{if } t \geq a \\ 0 & \text{if } t < a \end{cases}$$



$$y = u(t-a)$$

Consider the function :

$$y(t) = \begin{cases} f(t) & \text{if } t < a \\ g(t) & \text{if } a \leq t < b \\ h(t) & \text{if } t \geq b \end{cases}$$



We can express $y(t)$ in terms of step functions.

$$y(t) = f(t) + [g(t) - f(t)]u(t-a) + [h(t) - g(t)]u(t-b)$$

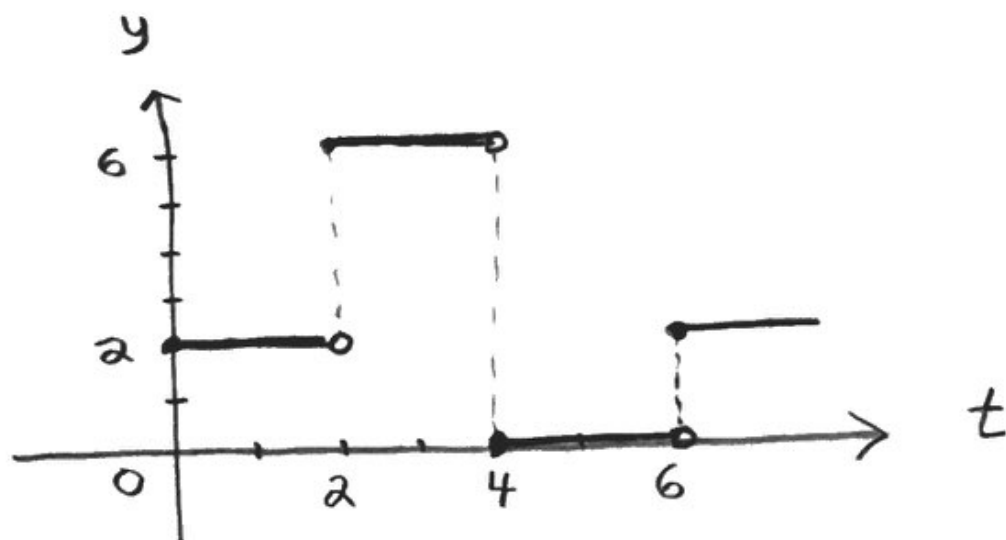
Note: if $t < a$, then: $\begin{cases} u(t-a) = 0 \\ u(t-b) = 0 \end{cases}$ Both are "off"

if $a \leq t < b$, then: $\begin{cases} u(t-a) = 1 \rightarrow \text{on} \\ u(t-b) = 0 \rightarrow \text{off} \end{cases}$

if $t \geq b$, then: $\begin{cases} u(t-a) = 1 \\ u(t-b) = 1 \end{cases}$ Both are "on".

ex: ④ Express using step functions

$$y(t) = \begin{cases} 2 & \text{if } 0 \leq t < 2 \\ 6 & \text{if } 2 \leq t < 4 \\ 0 & \text{if } 4 \leq t < 6 \\ 2 & \text{if } t \geq 6 \end{cases}$$



$$y(t) = 2 + 4u(t-2) - 6u(t-4) + 2u(t-6)$$

Annotations for the equation above:

- Value initially (points to 2)
- step up 4 units (points to $4u(t-2)$)
- at $t=2$ (points to $u(t-2)$)
- step down 6 units (points to $-6u(t-4)$)
- at $t=4$ (points to $u(t-4)$)
- step up 2 units (points to $2u(t-6)$)
- at $t=6$ (points to $u(t-6)$)

$$y(t) = 2 + 4u(t-2) - 6u(t-4) + 2u(t-6)$$

Note : we can also write $y(t)$ as

$$y(t) = \underline{\underline{2u(t)}} + 4u(t-2) - 6u(t-4) + 2u(t-6)$$

Since 2 and $2u(t)$ are
the same for $t \geq 0$.

We will now derive a useful formula
for $\mathcal{L}\{f(t-a)u(t-a)\}$.

By definition :

$$\mathcal{L}\{f(t-a)u(t-a)\} = \int_0^{\infty} e^{-st} f(t-a) u(t-a) dt$$

Since : $u(t-a) = \begin{cases} 1 & \text{if } t \geq a \\ 0 & \text{if } t < a \end{cases}$

$$\Rightarrow \mathcal{L}\{f(t-a)u(t-a)\} = \int_a^{\infty} e^{-st} f(t-a) dt$$

Let $u = t - a \rightarrow du = dt$

$$\begin{cases} t = \infty \rightarrow u = \infty \\ t = a \rightarrow u = 0 \end{cases}$$

$$\begin{aligned} \Rightarrow \mathcal{L}\{f(t-a)u(t-a)\} &= \int_0^{\infty} e^{-s(a+u)} f(u) du \\ &= e^{-as} \int_0^{\infty} e^{-su} f(u) du \\ &= e^{-as} F(s) \end{aligned}$$

This is called the 2nd shifting theorem.

If $F(s) = \mathcal{L}\{f(t)\}$ and $a > 0$,

then :

$$\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as} F(s)$$

We often use the inverse Laplace

version :

$$\mathcal{L}^{-1}\{e^{-as} F(s)\} = f(t-a)u(t-a)$$

$$\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as} F(s)$$

Two special cases

① If $f(t) = 1$ for all t
then $F(s) = \frac{1}{s}$

$$\Rightarrow \boxed{\mathcal{L}\{u(t-a)\} = \frac{e^{-as}}{s}}$$

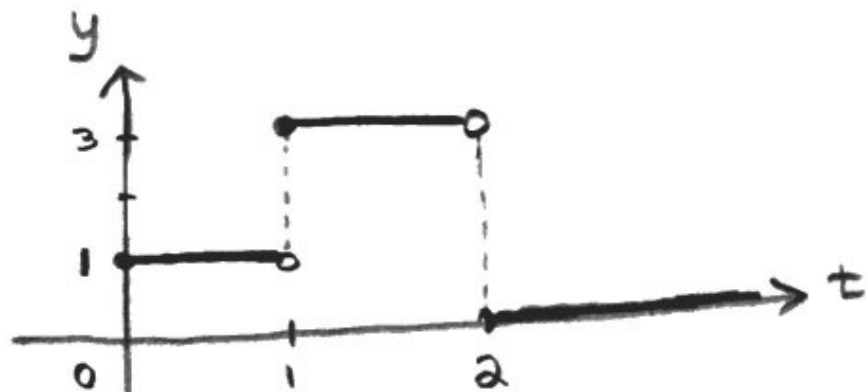
② If $g(t) = f(t-a)$
then $g(t+a) = f(t)$

$$\Rightarrow \mathcal{L}\{g(t)u(t-a)\} = e^{-as} F(s) = e^{-as} \mathcal{L}\{f(t)\}$$

$$\boxed{\mathcal{L}\{g(t)u(t-a)\} = e^{-as} \mathcal{L}\{g(t+a)\}}$$

ex: ⑤ Find $\mathcal{L}\{y(t)\}$ if

$$y(t) = \begin{cases} 1 & \text{if } 0 \leq t < 1 \\ 3 & \text{if } 1 \leq t < 2 \\ 0 & \text{if } t \geq 2 \end{cases}$$



$$y(t) = 1 + 2u(t-1) - 3u(t-2).$$

or

$$y(t) = u(t) + 2u(t-1) - 3u(t-2).$$

$$\mathcal{L}\{u(t-a)\} = \frac{e^{-as}}{s}$$

$$\Rightarrow \mathcal{L}\{y(t)\} = \frac{1}{s} + \frac{2e^{-s}}{s} - \frac{3e^{-2s}}{s}$$

ex: ⑥ Find $\mathcal{L}\{y(t)\}$ if

$$y(t) = \begin{cases} 0 & \text{if } 0 \leq t < 3 \\ t^2 & \text{if } t \geq 3 \end{cases}$$

observe that $y(t) = t^2 u(t-3)$.

We will use the formula :

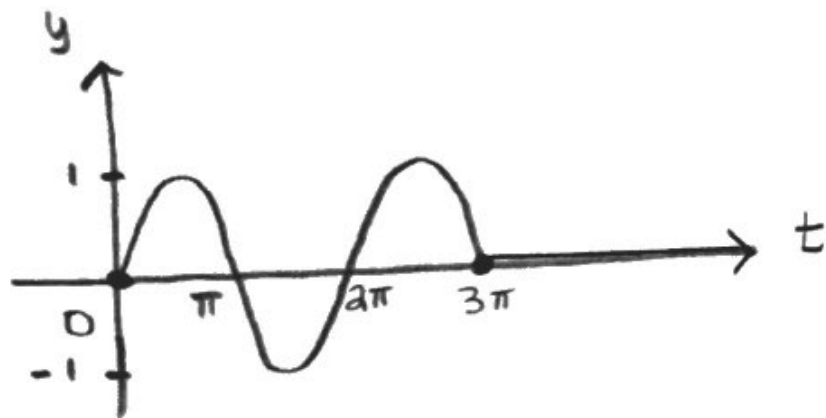
$$\mathcal{L}\{f(t) u(t-a)\} = e^{-as} \mathcal{L}\{f(t+a)\}.$$

$$\begin{aligned} \Rightarrow \mathcal{L}\{\underbrace{t^2}_{f(t)} u(t-3)\} &= e^{-3s} \mathcal{L}\{\underbrace{(t+3)^2}_{f(t+3)}\} \\ &= e^{-3s} \mathcal{L}\{t^2 + 6t + 9\} \end{aligned}$$

$$= \boxed{e^{-3s} \left(\frac{2}{s^3} + \frac{6}{s^2} + \frac{9}{s} \right)}$$

ex: ⑦ Find $\mathcal{L}\{y(t)\}$ if

$$y(t) = \begin{cases} \sin t & \text{if } 0 \leq t < 3\pi \\ 0 & \text{if } t \geq 3\pi \end{cases}$$



$$y(t) = \sin t - \sin t u(t-3\pi)$$

$$\mathcal{L}\{y(t)\} = \mathcal{L}\{\sin t\} - \mathcal{L}\{\sin t u(t-3\pi)\}$$

$$= \frac{1}{s^2+1} - e^{-3\pi s} \mathcal{L}\{\sin(t+3\pi)\}$$

$$= \frac{1}{s^2+1} - e^{-3\pi s} \mathcal{L}\{-\sin t\}$$

$$= \boxed{\frac{1}{s^2+1} + e^{-3\pi s} \left(\frac{1}{s^2+1} \right)}$$

ex: ⑧ Find $\mathcal{L}^{-1}\left\{\frac{e^{-3s}}{s-4}\right\}$.

Recall the formula:

$$\mathcal{L}^{-1}\{e^{-as}F(s)\} = f(t-a)u(t-a)$$

where: $\mathcal{L}^{-1}\{F(s)\} = f(t)$.

$$\mathcal{L}^{-1}\left\{\frac{1}{s-4}\right\} = e^{4t} \leftarrow f(t)$$

$$\Rightarrow \mathcal{L}^{-1}\left\{\frac{e^{-3s}}{s-4}\right\} = \mathcal{L}^{-1}\left\{e^{-3s}\left(\frac{1}{s-4}\right)\right\}$$

$\boxed{a=3}$

$$= \boxed{e^{4(t-3)} u(t-3)}$$

$f(t-a)$

$u(t-a)$

ex: ⑨ Find and simplify

$$\mathcal{L}^{-1} \left\{ e^{-\pi s} \left(\frac{s}{s^2 + 4} \right) \right\}.$$

Again, we use :

$$\mathcal{L}^{-1} \{ e^{-as} F(s) \} = f(t-a) u(t-a)$$

$$\mathcal{L}^{-1} \left\{ \frac{s}{s^2 + 4} \right\} = \cos(2t) \leftarrow f(t)$$

$$\Rightarrow \mathcal{L} \left\{ e^{-\pi s} \left(\frac{s}{s^2 + 4} \right) \right\} = f(t-\pi) u(t-\pi)$$

$$= \cos(2(t-\pi)) u(t-\pi)$$

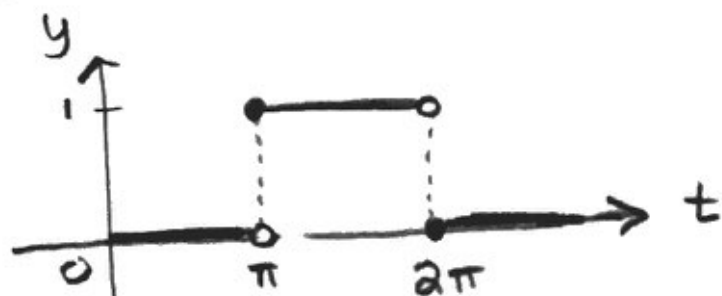
$$= \cos(2t - 2\pi) u(t-\pi)$$

$$= \boxed{\cos(2t) u(t-\pi)}$$

ex: (10) Solve the IVP

$$\begin{cases} y''(t) + y(t) = f(t) \\ y(0) = 0 \\ y'(0) = 1 \end{cases}$$

Where $f(t) = \begin{cases} 0 & \text{if } 0 \leq t < \pi \\ 1 & \text{if } \pi \leq t < 2\pi \\ 0 & \text{if } t \geq 2\pi \end{cases}$



$$f(t) = u(t - \pi) - u(t - 2\pi)$$

Take Laplace transform on both sides of the DE.

$$\mathcal{L}\{y''(t)\} + \mathcal{L}\{y(t)\} = \mathcal{L}\{f(t)\}$$

$$\left[s^2 Y(s) - s y(0) - y'(0) \right] + Y(s) = \frac{e^{-\pi s}}{s} - \frac{e^{-2\pi s}}{s}$$

$$\Rightarrow (s^2 + 1) Y(s) - s(0) - 1 = \frac{e^{-\pi s}}{s} - \frac{e^{-2\pi s}}{s}$$

$$\Rightarrow Y(s) = \frac{1}{s^2 + 1} + e^{-\pi s} \left(\frac{1}{s(s^2 + 1)} \right) - e^{-2\pi s} \left(\frac{1}{s(s^2 + 1)} \right)$$

$$\text{Let } W(s) = \frac{1}{s(s^2 + 1)}$$

$$\Rightarrow Y(s) = \frac{1}{s^2 + 1} + e^{-\pi s} W(s) - e^{-2\pi s} W(s)$$

$$\mathcal{L}^{-1}\{W(s)\} = w(t)$$

$$\Rightarrow y(t) = \mathcal{L}^{-1}\{Y(s)\}$$

$$\Rightarrow y(t) = \sin t + w(t - \pi) u(t - \pi) - w(t - 2\pi) u(t - 2\pi)$$

Let's use P.F. to find $w(t)$.

$$W(s) = \frac{1}{s(s^2+1)} = \frac{A}{s} + \frac{Bs+C}{s^2+1}$$

$$1 = A(s^2+1) + (Bs+C)s$$

$$1 = (A+B)s^2 + Cs + A$$

$$\Rightarrow \begin{cases} A+B=0 \rightarrow B=-1 \\ C=0 \\ A=1 \end{cases}$$

$$W(s) = \frac{1}{s} - \frac{s}{s^2+1}$$

$$w(t) = \mathcal{L}^{-1}\{W(s)\}$$

$$\Rightarrow w(t) = 1 - \cos t$$

$$y(t) = \sin t + (1 - \cos(t - \pi)) u(t - \pi) \dots$$

$$- (1 - \cos(t - 2\pi)) u(t - 2\pi)$$

$$\begin{cases} \cos(t - \pi) = -\cos t \\ \cos(t - 2\pi) = \cos t \end{cases}$$

$$\Rightarrow y(t) = \sin t + (1 + \cos t) u(t - \pi) \dots$$

$$- (1 - \cos t) u(t - 2\pi)$$

Alternatively, we can express $y(t)$ as a piecewise function.

$$y(t) = \begin{cases} \sin t & \text{if } 0 \leq t < \pi \\ \sin t + 1 + \cos t & \text{if } \pi \leq t < 2\pi \\ \sin t + 2\cos t & \text{if } t \geq 2\pi \end{cases}$$

Suppose that we want to evaluate $y(4)$.

We can plug $t = 4$ in $y(t)$.

$$y(4) = \sin(4) + (1 + \cos(4)) u(4 - \pi) - (1 - \cos(4)) u(4 - 2\pi)$$

$$u(4 - \pi) = +1 \quad \text{since } 4 - \pi > 0$$

$$u(4 - 2\pi) = 0 \quad \text{since } 4 - 2\pi < 0$$

$$\Rightarrow y(4) = \sin(4) + [1 + \cos(4)] (1) - [1 - \cos(4)] (0)$$

$$\Rightarrow \boxed{y(4) = \sin(4) + 1 + \cos(4)}$$

We can also get $y(4)$ from the piecewise function by using the middle entry since

$$\pi \leq 4 < 2\pi.$$

ex: ② Consider an LR-circuit

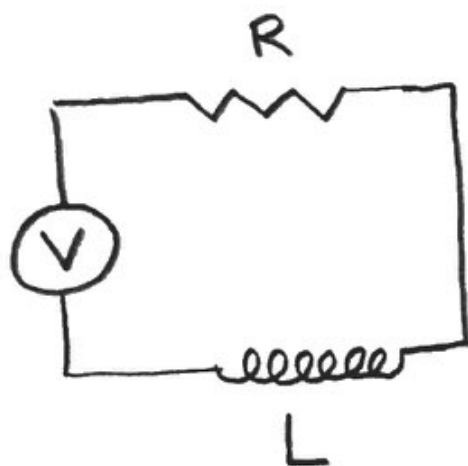
with $L = 1 \text{ H}$, $R = 6 \Omega$

For simplicity

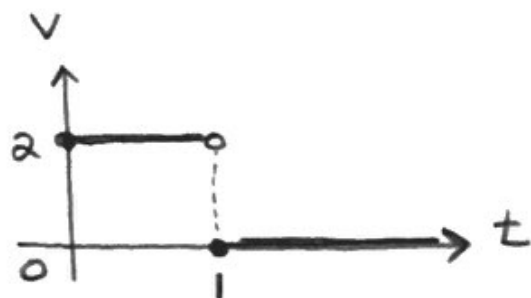
and a voltage source (in Volts)

$$V(t) = \begin{cases} 2 & \text{if } 0 \leq t < 1 \\ 0 & \text{if } t \geq 1 \end{cases}$$

where t is time (in seconds).



- (a) Find the current $i(t)$ if $i(0) = 0$.
- (b) Find the current at $t = 2$ seconds.



$$V(t) = 2 - 2u(t-1)$$

The DE for this circuit is

$$L \frac{di}{dt} + R i(t) = V(t)$$

$$\Rightarrow \frac{di}{dt} + 6 i(t) = 2 - 2u(t-1)$$

Take Laplace on both sides to get:

$$\left[s I(s) - i(0) \right] + 6 I(s) = \frac{2}{s} - \frac{2e^{-s}}{s}$$

since $i(0) = 0$

$$\Rightarrow (s+6) I(s) = \frac{2}{s} - \frac{2e^{-s}}{s}$$

$$\Rightarrow I(s) = \frac{2}{s(s+6)} - e^{-s} \left(\frac{2}{s(s+6)} \right)$$

$$\text{Let } W(s) = \frac{2}{s(s+6)}$$

$$\Rightarrow I(s) = W(s) - e^{-s} W(s)$$

$$i(t) = \mathcal{L}^{-1}\{I(s)\}$$

$$i(t) = w(t) - w(t-1)u(t-1)$$

Now let's use P.F. to find $w(t)$.

$$W(s) = \frac{2}{s(s+6)} = \frac{A}{s} + \frac{B}{s+6}$$

$$\Rightarrow \begin{cases} A = 1/3 \\ B = -1/3 \end{cases}$$

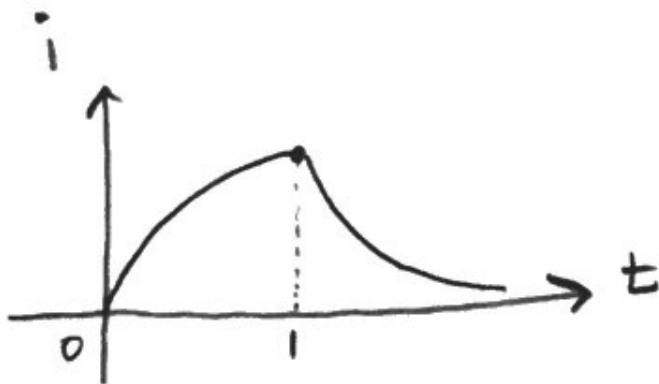
$$\Rightarrow w(t) = \frac{1}{3} - \frac{1}{3} e^{-6t}$$

(a) The current is then:

$$i(t) = \left(\frac{1}{3} - \frac{1}{3} e^{-6t} \right) - \left(\frac{1}{3} - \frac{1}{3} e^{-6(t-1)} \right) u(t-1)$$

or

$$i(t) = \begin{cases} \frac{1}{3} - \frac{1}{3} e^{-6t} & \text{if } 0 \leq t < 1 \\ -\frac{1}{3} e^{-6t} + \frac{1}{3} e^{-6(t-1)} & \text{if } t \geq 1 \end{cases}$$



(b) @ $t = 2$ seconds, the current is

$$i(2) = -\frac{1}{3} e^{-12} + \frac{1}{3} e^{-6}$$

$$= 8.24 \times 10^{-4} \text{ A}$$

$$= \boxed{824 \mu\text{A}}$$